

XAllergen

(eXplainable Allergenicity Prediction of Proteins)

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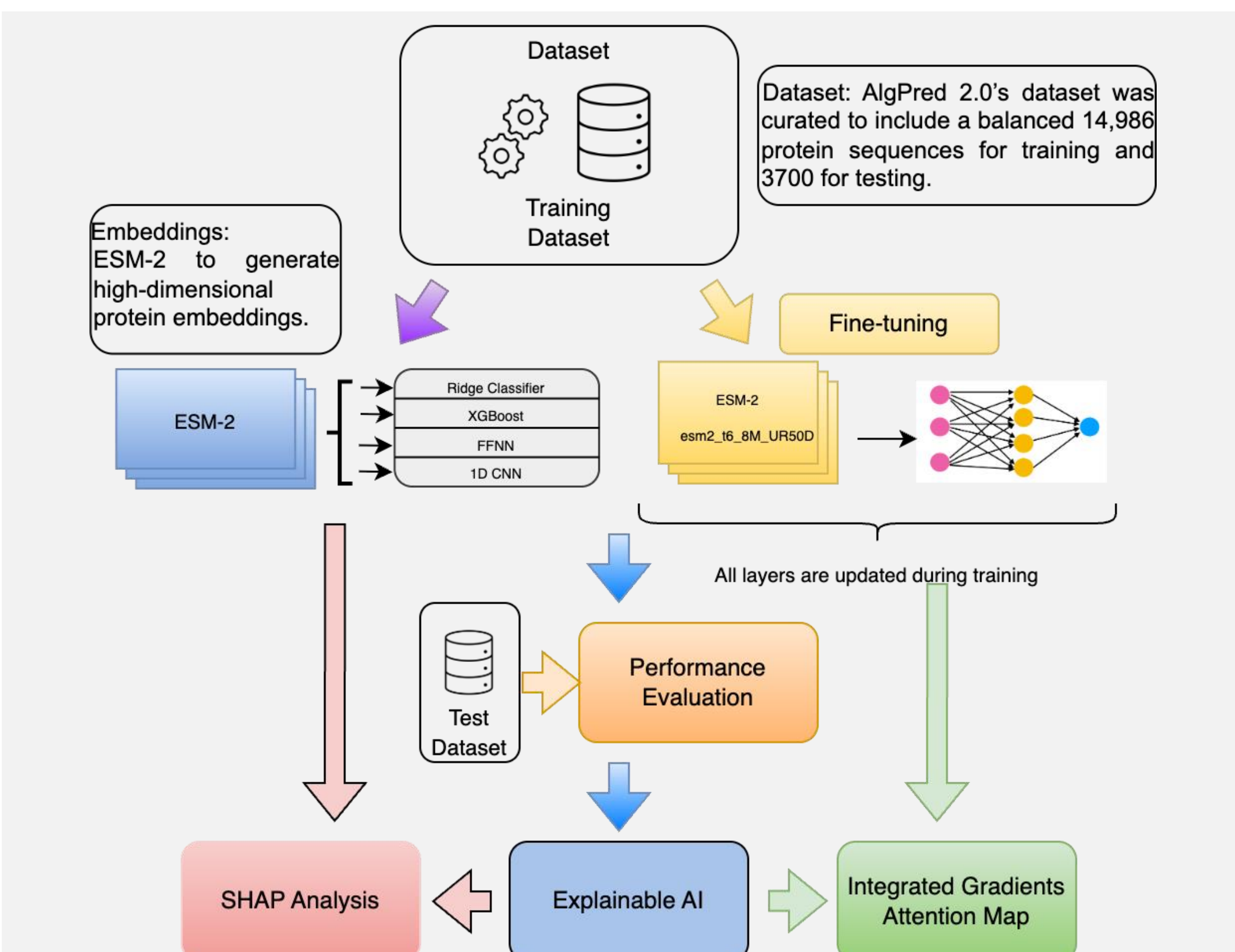
Scan to view the code in Github:
github.com/Jeffateth/XAllergen



1 Introduction

- Problem:**
 - Allergies are affecting 1/3 of the population.¹
 - Experimental allergenicity classification is time-consuming and costly.
 - Most existing allergenicity prediction models lack interpretability, limiting their usefulness in scientific and clinical settings.
- Solution:**
Develop **XAllergen**, a sequence-based explainable AI tool that enables fast and interpretable prediction of protein allergenicity using ESM-2 embeddings and attention-based deep learning on an interactive website.

2 Method Overview



Model Architecture:

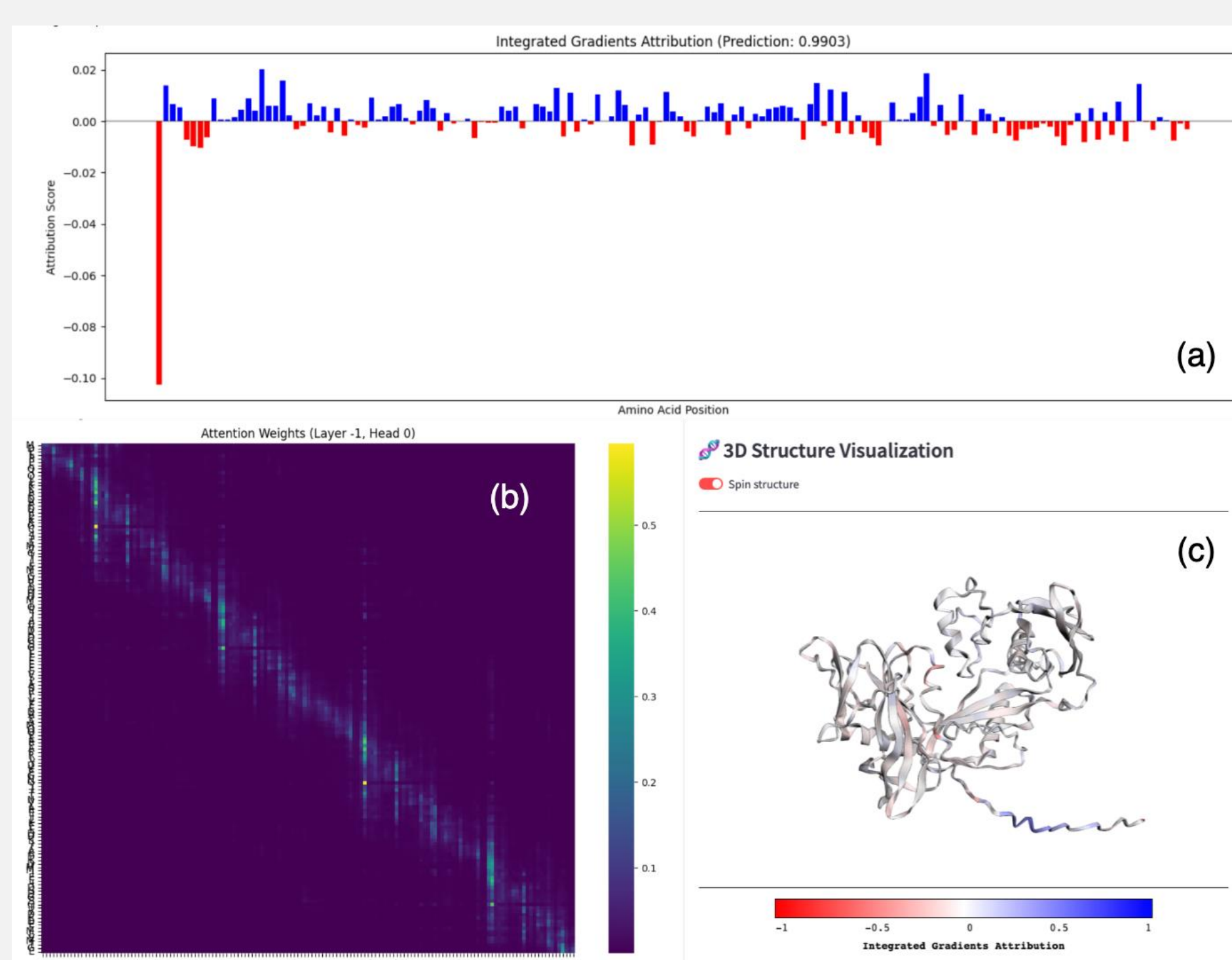
Trained XGBoost, Ridge Classifier, Neural Networks (MLP, CNN), and a fine-tuned ESM-2 classifier.

Interpretability:

- Applied Integrated Gradients² to trace model predictions back to amino acids.
- Visualized attention maps from ESM-2's final transformer layer.
- Created 3D attribution maps (py3Dmol) using ESMFold-predicted structures + DSSP-derived features.

Deployment:

Built an interactive Streamlit web app to allow users to upload sequences, receive predictions, and visualize attribution maps in real-time.



4 Results and Discussion

We compared multiple models trained on ESM-2 embeddings and tested on the same independent dataset. The best MCC performance was achieved by the XGBoost ESM-2 1280dim (with accuracy = 88%, ROC-AUC = 98%). However, for deployment, we selected the fine-tuned ESM-2 model due to its balance between interpretability and performance (accuracy = 84%, ROC-AUC = 92%). As shown in the chart below, our model outperformed the dummy model (50%) across all metrics, all with high specificity, reliably identifying true allergens when making positive predictions. Attempts to fuse 3D structural features with ESM embeddings yielded limited improvements, highlighting challenges in feature integration.

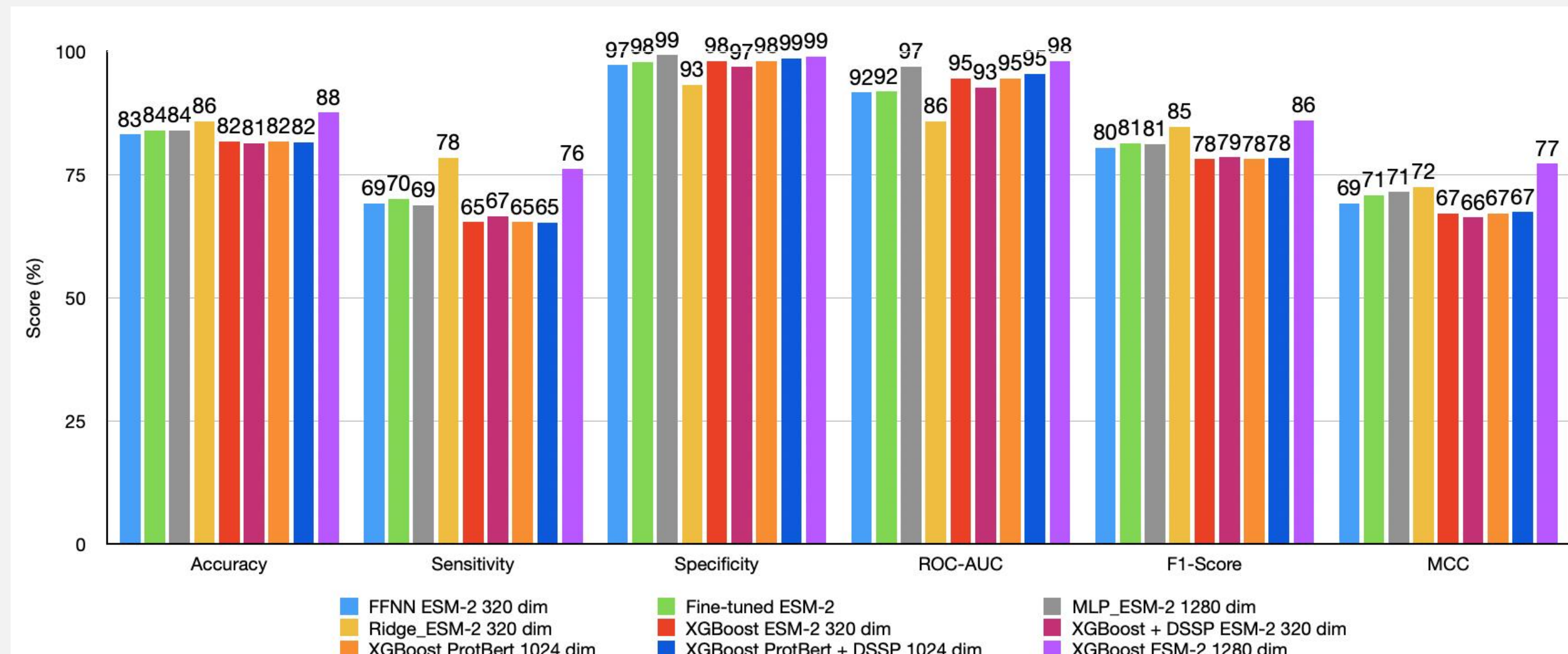


Figure 1: Comparison between different models using different performance metrics: accuracy, sensitivity, specificity, ROC-AUC, F1-Score and MCC

5 Conclusion

- We developed XAllergen, an explainable AI tool for protein allergenicity prediction based on ESM-2 embeddings.
- The model showed strong performance on a curated balanced independent test set with robust performance and interpretability.
- An interactive Streamlit interface enables real-time predictions and visualization, making allergen assessment more accessible.
- Created ~20,000 first-hand predicted 3D PDB structures using ESMFold.
- Future work:**
 - incorporate 3D structural features to improve sensitivity
 - improving real-time 3D interpretation support
 - extend sequence length beyond 1022 residues
 - enrich prediction results with biological contexts
 - identify occurring allergenic patterns across proteins to study which protein family is prone to cause allergy

References

- Allergien – Globale Patientenplattform für Allergien und Atemwege. Global Allergy & Airways Patient Platform. <https://de.gaapp.org/diseases/allergies>.
- Mukund Sundararajan, Ankur Taly, and Qiqi Yan. "Axiomatic Attribution for Deep Networks". In: Proceedings of the 34th International Conference on Machine Learning.